八十三年第一次專門職業及技術人員 律師、會計師 檢覈筆試試題
建築師、技師

全一張
(正面)

類　　科：冷凍空調工程技師
科　　目：空　調　工　程
考試時間：二 小 時

一、在空調工程中，用於濾清空氣，有那幾種方法，請簡述其原理。（15分）

二、若空氣的乾球溫度爲15°C，體積爲150m³，試求此空氣的標準空氣(Standard Air)的體積。（15分）

三、某室內空氣的壓力、溫度、相對濕度分別爲100kpa、30°C、70%，試求此空氣的濕度比（Humidity ratio)。（15分）

四、若空氣的相對濕度爲80%，乾球溫度爲20°C，今欲將此空氣加熱至30°C，試求每一仟克的乾空氣應加熱量多少？（25分）
設：大氣壓力爲101.3kpa，空氣在加熱過程中爲穩定流動。

五、一空調用的冷却水塔，欲將45°C的水冷至25°C，水的流量率爲110,000仟克／小時，空氣自水塔底部流入，其溫度爲20°C，相對濕度爲55%，空氣自水塔頂部流出，其溫度爲40°C，相對濕度爲95%，若大氣壓力爲92.0kpa，試求此水塔每小時需多少的冷却空氣質量？（30分）

參考附表：

**Table B-1** SATURATED WATER—TEMPERATURE TABLE

| TEMP., °C, $T$ | PRESS., kPa $P$ | SPECIFIC VOLUME, m³/kg SAT. LIQUID, $v_f$ | SAT. VAPOR, $v_g$ | INTERNAL ENERGY, kJ/kg SAT. LIQUID, $u_f$ | EVAP., $u_{fg}$ | SAT. VAPOR, $u_g$ | ENTHALPY, kJ/kg SAT. LIQUID, $h_f$ | EVAP., $h_{fg}$ | SAT. VAPOR, $h_g$ | ENTROPY, kJ/kg·K SAT. LIQUID, $s_f$ | EVAP., $s_{fg}$ | SAT. VAPOR, $s_g$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.01 | 0.6113 | 0.001 000 | 206.14 | .00 | 2375.3 | 2375.3 | .01 | 2501.3 | 2501.4 | .0000 | 9.1562 | 9.1562 |
| 5 | 0.8721 | 0.001 000 | 147.12 | 20.97 | 2361.3 | 2382.3 | 20.98 | 2489.6 | 2510.6 | .0761 | 8.9496 | 9.0257 |
| 10 | 1.2276 | 0.001 000 | 106.38 | 42.00 | 2347.2 | 2389.2 | 42.01 | 2477.7 | 2519.8 | .1510 | 8.7498 | 8.9008 |
| 15 | 1.7051 | 0.001 001 | 77.93 | 62.99 | 2333.1 | 2396.1 | 62.99 | 2465.9 | 2528.9 | .2245 | 8.5569 | 8.7814 |
| 20 | 2.339 | 0.001 002 | 57.79 | 83.95 | 2319.0 | 2402.9 | 83.96 | 2454.1 | 2538.1 | .2966 | 8.3706 | 8.6672 |
| 25 | 3.169 | 0.001 003 | 43.36 | 104.88 | 2304.9 | 2409.8 | 104.89 | 2442.3 | 2547.2 | .3674 | 8.1905 | 8.5580 |
| 30 | 4.246 | 0.001 004 | 32.89 | 125.78 | 2290.8 | 2416.6 | 125.79 | 2430.5 | 2556.3 | .4369 | 8.0164 | 8.4533 |
| 35 | 5.628 | 0.001 006 | 25.22 | 146.67 | 2276.7 | 2423.4 | 146.68 | 2418.6 | 2565.3 | .5053 | 7.8478 | 8.3531 |
| 40 | 7.384 | 0.001 008 | 19.52 | 167.56 | 2262.6 | 2430.1 | 167.57 | 2406.7 | 2574.3 | .5725 | 7.6845 | 8.2570 |
| 45 | 9.593 | 0.001 010 | 15.26 | 188.44 | 2248.4 | 2436.8 | 188.45 | 2394.8 | 2583.2 | .6387 | 7.5261 | 8.1648 |
| 50 | 12.349 | 0.001 012 | 12.03 | 209.32 | 2234.2 | 2443.5 | 209.33 | 2382.7 | 2592.1 | .7038 | 7.3725 | 8.0763 |
| 55 | 15.758 | 0.001 015 | 9.568 | 230.21 | 2219.9 | 2450.1 | 230.23 | 2370.7 | 2600.9 | .7679 | 7.2234 | 7.9913 |
| 60 | 19.940 | 0.001 017 | 7.671 | 251.11 | 2205.5 | 2456.6 | 251.13 | 2358.5 | 2609.6 | .8312 | 7.0784 | 7.9096 |
| 65 | 25.03 | 0.001 020 | 6.197 | 272.02 | 2191.1 | 2463.1 | 272.06 | 2346.2 | 2618.3 | .8935 | 6.9375 | 7.8310 |
| 70 | 31.19 | 0.001 023 | 5.042 | 292.95 | 2176.6 | 2469.6 | 292.98 | 2333.8 | 2626.8 | .9549 | 6.8004 | 7.7553 |
| 75 | 38.58 | 0.001 026 | 4.131 | 313.90 | 2162.0 | 2475.9 | 313.93 | 2321.4 | 2635.3 | 1.0155 | 6.6669 | 7.6824 |
| 80 | 47.39 | 0.001 029 | 3.407 | 334.86 | 2147.4 | 2482.2 | 334.91 | 2308.8 | 2643.7 | 1.0753 | 6.5369 | 7.6122 |
| 85 | 57.83 | 0.001 033 | 2.828 | 355.84 | 2132.6 | 2488.4 | 355.90 | 2296.0 | 2651.9 | 1.1343 | 6.4102 | 7.5445 |
| 90 | 70.14 | 0.001 036 | 2.361 | 376.85 | 2117.7 | 2494.5 | 376.92 | 2283.2 | 2660.1 | 1.1925 | 6.2866 | 7.4791 |
| 95 | 84.55 | 0.001 040 | 1.982 | 397.88 | 2102.7 | 2500.6 | 397.96 | 2270.2 | 2668.1 | 1.2500 | 6.1659 | 7.4159 |

| TEMP., °C, $T$ | PRESS., MPa $P$ | SPECIFIC VOLUME, m³/kg SAT. LIQUID, $v_f$ | SAT. VAPOR, $v_g$ | INTERNAL ENERGY, kJ/kg SAT. LIQUID, $u_f$ | EVAP., $u_{fg}$ | SAT. VAPOR, $u_g$ | ENTHALPY, kJ/kg SAT. LIQUID, $h_f$ | EVAP., $h_{fg}$ | SAT. VAPOR, $h_g$ | ENTROPY, kJ/kg·K SAT. LIQUID, $s_f$ | EVAP., $s_{fg}$ | SAT. VAPOR, $s_g$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 100 | 0.101 35 | 0.001 044 | 1.6729 | 418.94 | 2087.6 | 2506.5 | 419.04 | 2257.0 | 2676.1 | 1.3069 | 6.0480 | 7.3549 |
| 105 | 0.120 82 | 0.001 048 | 1.4194 | 440.02 | 2072.3 | 2512.4 | 440.15 | 2243.7 | 2683.8 | 1.3630 | 5.9328 | 7.2958 |
| 110 | 0.143 27 | 0.001 052 | 1.2102 | 461.14 | 2057.0 | 2518.1 | 461.30 | 2230.2 | 2691.5 | 1.4185 | 5.8202 | 7.2387 |
| 115 | 0.169 06 | 0.001 056 | 1.0366 | 482.30 | 2041.4 | 2523.7 | 482.48 | 2216.5 | 2699.0 | 1.4734 | 5.7100 | 7.1833 |
| 120 | 0.198 53 | 0.001 060 | 0.8919 | 503.50 | 2025.8 | 2529.3 | 503.71 | 2202.6 | 2706.3 | 1.5276 | 5.6020 | 7.1296 |
| 125 | 0.2321 | 0.001 065 | 0.7706 | 524.74 | 2009.9 | 2534.6 | 524.99 | 2188.5 | 2713.5 | 1.5813 | 5.4962 | 7.0775 |
| 130 | 0.2701 | 0.001 070 | 0.6685 | 546.02 | 1993.9 | 2539.9 | 546.31 | 2174.2 | 2720.5 | 1.6344 | 5.3925 | 7.0269 |
| 135 | 0.3130 | 0.001 075 | 0.5822 | 567.35 | 1977.7 | 2545.0 | 567.69 | 2159.6 | 2727.3 | 1.6870 | 5.2907 | 6.9777 |
| 140 | 0.3613 | 0.001 080 | 0.5089 | 588.74 | 1961.3 | 2550.0 | 589.13 | 2144.7 | 2733.9 | 1.7391 | 5.1908 | 6.9299 |
| 145 | 0.4154 | 0.001 085 | 0.4463 | 610.18 | 1944.7 | 2554.9 | 610.63 | 2129.6 | 2740.3 | 1.7907 | 5.0926 | 6.8833 |

(請接背面)

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科　目：空　調　工　程

**Table D-1** IDEAL-GAS PROPERTIES OF AIR (MASS BASIS)

| $T$, K | $h$, kJ/kg | $u$, kJ/kg | $s^\circ$, kJ/kg·K | $P_r$ | $v_r$ |
|---|---|---|---|---|---|
| 100 | 99.76 | 71.06 | 5.6008 | 0.02990 | 2230 |
| 110 | 109.77 | 78.20 | 5.6963 | 0.04171 | 1758.4 |
| 120 | 119.79 | 85.34 | 5.7836 | 0.05652 | 1415.7 |
| 130 | 129.81 | 92.51 | 5.8638 | 0.07474 | 1159.8 |
| 140 | 139.84 | 99.67 | 5.9380 | 0.09681 | 964.2 |
| 150 | 149.86 | 106.81 | 6.0071 | 0.12318 | 812.0 |
| 160 | 159.87 | 113.95 | 6.0718 | 0.15431 | 691.4 |
| 170 | 169.89 | 121.11 | 6.1326 | 0.19068 | 594.5 |
| 180 | 179.92 | 128.28 | 6.1898 | 0.23279 | 515.6 |
| 190 | 189.94 | 135.40 | 6.2440 | 0.28114 | 450.6 |
| 200 | 199.96 | 142.56 | 6.2953 | 0.3363 | 396.6 |
| 210 | 209.97 | 149.70 | 6.3442 | 0.3987 | 351.2 |
| 220 | 219.99 | 156.84 | 6.3908 | 0.4690 | 312.8 |
| 230 | 230.01 | 163.98 | 6.4354 | 0.5477 | 280.0 |
| 240 | 240.03 | 171.15 | 6.4780 | 0.6355 | 251.8 |
| 250 | 250.05 | 178.29 | 6.5190 | 0.7329 | 227.45 |
| 260 | 260.09 | 185.45 | 6.5582 | 0.8405 | 206.26 |
| 270 | 270.12 | 192.59 | 6.5961 | 0.9590 | 187.74 |
| 280 | 280.14 | 199.78 | 6.6326 | 1.0889 | 171.45 |
| 290 | 290.17 | 206.92 | 6.6678 | 1.2311 | 157.07 |
| 300 | 300.19 | 214.09 | 6.7018 | 1.3860 | 144.32 |
| 310 | 310.24 | 221.27 | 6.7348 | 1.5546 | 132.96 |
| 320 | 320.29 | 228.45 | 6.7667 | 1.7375 | 122.81 |
| 330 | 330.34 | 235.65 | 6.7976 | 1.9352 | 113.70 |
| 340 | 340.43 | 242.86 | 6.8277 | 2.149 | 105.51 |
| 350 | 350.48 | 250.05 | 6.8569 | 2.379 | 98.11 |
| 360 | 360.58 | 257.23 | 6.8852 | 2.626 | 91.40 |
| 370 | 370.67 | 264.47 | 6.9129 | 2.892 | 85.31 |
| 380 | 380.77 | 271.72 | 6.9399 | 3.176 | 79.77 |
| 390 | 390.88 | 278.96 | 6.9661 | 3.481 | 74.71 |
| 400 | 400.98 | 286.19 | 6.9917 | 3.806 | 70.07 |
| 410 | 411.12 | 293.45 | 7.0167 | 4.153 | 65.83 |
| 420 | 421.26 | 300.73 | 7.0412 | 4.522 | 61.93 |
| 430 | 431.43 | 308.03 | 7.0651 | 4.915 | 58.34 |
| 440 | 441.61 | 315.34 | 7.0885 | 5.332 | 55.02 |
| 450 | 451.83 | 322.66 | 7.1114 | 5.775 | 51.96 |
| 460 | 462.01 | 329.99 | 7.1338 | 6.245 | 49.11 |
| 470 | 472.25 | 337.34 | 7.1558 | 6.742 | 46.48 |
| 480 | 482.48 | 344.74 | 7.1774 | 7.268 | 44.04 |
| 490 | 492.74 | 352.11 | 7.1985 | 7.824 | 41.76 |
| 500 | 503.02 | 359.53 | 7.2193 | 8.411 | 39.64 |
| 510 | 513.32 | 366.97 | 7.2397 | 9.031 | 37.65 |
| 520 | 523.63 | 374.39 | 7.2598 | 9.684 | 35.80 |
| 530 | 533.98 | 381.88 | 7.2795 | 10.372 | 34.07 |
| 540 | 544.35 | 389.40 | 7.2989 | 11.097 | 32.45 |
| 550 | 554.75 | 396.89 | 7.3179 | 11.858 | 30.92 |

SOURCE: Adapted to SI units from *Gas Tables* by Joseph H. Keenan and Joseph Kaye. Copyright by John Wiley & Sons, Inc., 1948.

**Table D-9** ZERO-PRESSURE SPECIFIC HEATS AND $k$ AT 27°C FOR COMMON GASES

| GAS | CHEMICAL FORMULA | MOLECULAR WEIGHT | $R$, kJ/kg·K | $c_{p0}$, kJ/kg·K | $c_{v0}$, kJ/kg·K | $k$ |
|---|---|---|---|---|---|---|
| Air | — | 28.97 | 0.287 00 | 1.005 | 0.718 | 1.400 |
| Argon | Ar | 39.948 | 0.208 13 | 0.5203 | 0.3122 | 1.667 |
| Butane | $C_4H_{10}$ | 58.124 | 0.143 04 | 1.7164 | 1.5734 | 1.091 |
| Carbon dioxide | $CO_2$ | 44.011 | 0.188 92 | 0.846 | 0.657 | 1.289 |
| Carbon monoxide | CO | 28.011 | 0.296 83 | 1.040 | 0.744 | 1.400 |
| Ethane | $C_2H_6$ | 30.070 | 0.276 50 | 1.7662 | 1.4897 | 1.186 |
| Ethylene | $C_2H_4$ | 28.054 | 0.296 37 | 1.5482 | 1.2518 | 1.237 |
| Helium | He | 4.003 | 2.077 03 | 5.1926 | 3.1156 | 1.667 |
| Hydrogen | $H_2$ | 2.016 | 4.124 18 | 14.307 | 10.183 | 1.405 |
| Methane | $CH_4$ | 16.043 | 0.518 35 | 2.2537 | 1.7354 | 1.299 |
| Neon | Ne | 20.183 | 0.411 95 | 1.0299 | 0.6179 | 1.667 |
| Nitrogen | $N_2$ | 28.013 | 0.296 80 | 1.039 | 0.743 | 1.400 |
| Octane | $C_8H_{18}$ | 114.23 | 0.072 79 | 1.7113 | 1.6385 | 1.044 |
| Oxygen | $O_2$ | 31.999 | 0.259 83 | 0.918 | 0.658 | 1.395 |
| Propane | $C_3H_8$ | 44.097 | 0.188 55 | 1.6794 | 1.4909 | 1.126 |
| Steam | $H_2O$ | 18.015 | 0.461 52 | 1.8723 | 1.4108 | 1.327 |

SOURCE: Gordon J. Van Wylen and Richard E. Sonntag, *Fundamentals of Classical Thermodynamics*, SI Version, 2nd Ed, Revised Printing, Copyright © 1976, 1978 by John Wiley & Sons, Inc. Table A-8, pp. 683.